Demystifying MHz		
Processor	**Core speed**	**Bus Speed**
Athlon XP 1500+	1333 MHz	266 MHz
Athlon XP 1600+	1400 MHz	266 MHz
Athlon XP 1700+	1467 MHz	266 MHz
Athlon XP 1800+	1533 MHz	266 MHz
Athlon XP 1900+	1600 MHz	266 MHz
Athlon XP 2000+	1667 MHz	266 MHz
Athlon XP 2100+	1733 MHz	266 MHz
Athlon XP 2200+	1800 MHz	266 MHz
Athlon XP 2400+	2000 MHz	266 MHz
Athlon XP 2600+	2133 MHz	266 MHz
Athlon XP 2700+	2166 MHz	333 MHz
Athlon XP 2800+	2250 MHz	333 MHz

Unlike Intel, which measures CPU speed in megahertz (MHz), AMD uses a system called PR rating. This table shows the corresponding MHz ratings for different flavours of Athlon XPs

(by more than Rs 2,000) DDR-based Intel board comes across as a better buy.

The SiS chipset-based Mercury motherboard surprised us with its great performance number (71.60). Considering its price, Rs 3,600, the Mercury KOB650GL is an outstanding buy under this application category.

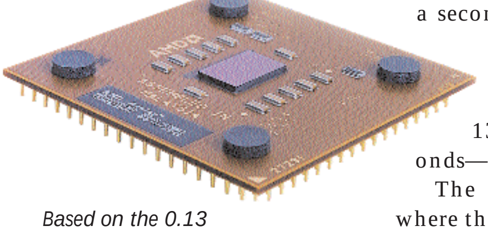
In the AMD camp, MSI stole ASUS' thunder by a decent margin. Scoring 76.30 STTI, the MSI KT4-Ultra is the best board that an Athlon can hope for: feature-rich (onboard six-channel sound, onboard FireWire, USB 2.0, Ethernet, RAID, Bluetooth, Serial ATA) and good performing. Only the DFI AD77-infinity came close to touching the MSI variant both in performance (score of 72.87 STTI) and in features. But when price is taken into consideration, at Rs 6,940 the MSI silences all and comes across as a great buy for this application area.

The best processors for gaming and entertainment

Although dominated by AMD processors, this application area saw the latest silicon square from Intel, the P4 3.06 GHz, annihilate its competition. This 3.06 GHz chip features the much-awaited HyperThreading feature, while the rest of the architecture is identical to other P4 chips. (*To learn more about Hyper-Threading, refer to the box, 'Virtual juggling: A look at Hyper-Threading'*).

This Intel chip was able to encode our test audio file of 51 MB to MP3 in just 17 seconds. In the video encoding test, it encoded the test file in 1 minute 28 seconds. Impressed? As good as this processor is, the 3.06 GHz is frequently bested by the Athlon XP 2800+ as you will soon see.

AMD Athlon XP 2800+, the Thoroughbred, managed to give the P4 3.06 GHz a serious run in every benchmark. Bear in mind, that the 2800+ runs approximately 800 MHz slower than the P4 3.06 GHz. Interestingly, it still manages to outperform the P4 in CPU-hungry tests such as audio and video encoding and even in our POVRAY test. The 2800+ finished encoding the video file in a second less than the P4 3.06 GHz; while it ripped the audio file into MP3 in 13 seconds 4 seconds—faster than the P4.



Based on the 0.13 micron Thoroughbred core, the 2800+ is the quickest of the AMD breed

The gaming tests is where the two part ways by a margin: in 3DMark 2001 the P4 scores a healthy 12,981 3D Marks, a good 500 points